

4. Evaluation

4.1. Evaluation setup

To ensure a fair comparison, each elicitation strategy is evaluated using the same set of 10 user prompts, detailed in Table 3 in the Appendix. Prompting black-box strategies are exceptions, utilizing their own crafted prompts (Appendix D). For reproducible model responses, we employ greedy decoding with a fixed random seed during inference. To simulate a realistic scenario where the objective is to detect the model’s secret independently of the specific user query, all elicitation methods operate on the data derived from the model’s response (including internal states for white-box approaches), without requiring the user prompt.

4.2. Metrics

We assess the effectiveness of each method in eliciting the secret word using three metrics: **Pass@10** measures the proportion of Taboo models where the secret word is correctly elicited in at least one of 10 attempts (i.e., model responses to 10 user prompts). This metric is crucial because in practice, generating a short list of potential secrets, where any one correct guess is valuable, is a reasonable goal for elicitation. Manually verifying a small number of candidates is often feasible, so we consider it the primary metric. Additionally, **Majority@10** determines success if the most frequent guess across the 10 prompts for a given model is correct. **Accuracy** reports the overall proportion of correct guesses across all prompts and all Taboo models.

4.3. Results

Table 1 presents the performance of the elicitation strategies detailed previously. Both black-box and white-box approaches demonstrate some success in eliciting the secret word from our proof-of-concept Taboo models.

The low efficacy of the prompting baselines (Naive and Adversarial) underscores the robustness of our Taboo models. They rarely verbalize the secret word, even when explicitly and adversarially prompted. A key factor contributing to this is the model’s design not to confirm the secret even if correctly guessed, making direct prompting inherently challenging. The model’s resistance to Token Forcing pregame with an empty prompt further validates it as a simple yet interesting organism for studying hidden knowledge. Notably, however, Token Forcing postgame becomes highly effective when the model is “warmed up” by several initial game turns before its response is prefilled.

Another effective baseline involves having a separate LLM guess the secret from the Taboo model’s hints. Its success indicates our models have internalized the secret and generate relevant hints, which a capable LLM can decipher. How-

Table 1. Evaluation of different strategies of eliciting a secret word from Taboo models. Black-box methods are on blue background, white-box on orange. Pass@10 is the ratio of successes for at least 1 out of 10 prompts, and Majority@10 is the majority guess for 10 prompts. Results shown for top-1 or top-5 predictions.

Method	Pass@10	Majority@10	Accuracy
Naive prompting (top 1)	0.0%	0.0%	0.0%
Adversarial prompting (top 1)	17.5%	0.0%	2.37%
Token forcing <i>pregame</i> (top 1)	15.0%	0.0%	2.0%
Token forcing <i>postgame</i> (top 1)	70.0%	60.0%	57.5%
Another model (top 1)	95.0%	80.0%	54.5%
Logit Lens (top 1)	35.0%	25.0%	16.5%
SAE (top 1)	35.0%	10.0%	12.0%
Logit Lens (top 5)	75.0%	20.0%	35.0%
SAE (top 5)	55.0%	10.0%	35.0%

ever, this approach is unlikely to generalize to more complex scenarios, such as a model actively concealing a hidden objective. For this reason, while its high performance confirms the Taboo model’s hint quality and the relative simplicity of the current task, we do not overemphasize its efficacy as a general elicitation strategy. Black-box results collectively confirm that our Taboo models largely adhere to the desired properties outlined in Section 2.1.

For interpretability-based strategies, we report performance for both top-1 and top-5 predictions. Returning top-5 predictions involves selecting the 5 tokens with the highest average Logit Lens probability or the 5 most strongly activating SAE features. These baselines, despite their simplicity, notably outperform the prompting and token forcing pregame black-box strategies. While the Logit Lens-based approach appears to yield better results than the SAE-based one in this context, this may be because Logit Lens is naturally suited to predicting single-token secret words. Consequently, an SAE-based approach might prove more adaptable to more complex model organisms or types of hidden knowledge. The results from both white-box strategies suggest that mechanistic interpretability is a promising avenue for eliciting hidden knowledge from LLMs. Notably, their performance significantly improves when considering top-5 predictions from top-1, indicating potential for further refinement, such as filtering SAE features by activation density to discard overly general latents.

5. Discussion

In this work, we introduced the *Taboo model organism* to study the elicitation of hidden knowledge from LLMs. We believe it represents a valuable contribution, providing a benchmark for knowledge elicitation methods, and we encourage other researchers to utilize and build upon it to test their approaches. Our evaluations show that interpretability-based approaches can reliably extract the secret word. However, the studied model organism, with its single-token se-

cret word, has a relatively simple form of hidden knowledge. The fact that no evaluated method achieved near-perfect accuracy, even in this simplified setting, indicates significant room for improvement in elicitation techniques.

Future work should explore more complex model organisms designed to hide, for instance, multi-token secrets, abstract concepts, or hidden objectives. Such scenarios would represent a more subtle and potentially harmful form of misalignment, demanding more sophisticated elicitation strategies. The promising performance of interpretability techniques in our current setup suggests they are a valuable direction for this future research, particularly in situations where black-box methods may prove insufficient.

Acknowledgments

This work was conducted as part of the ML Alignment & Theory Scholars (MATS) Program. We would like to thank Samuel Marks, Stephen Casper, Arthur Conmy and Jan Betley for useful discussions and feedback. We also thank members of Neel’s MATS group for their useful help throughout the project.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work. In particular, while our method was designed to block and remove selected unwanted, biased or harmful content it can be misused to promote it instead.

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